GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points · two class periods **WORKED ANSWERS**

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors *10 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

k = 2.5

copy ÷ original

Scoring: 1 point.

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.4

copy ÷ original; the copy is smaller, so k is under 1 — 2.5, 4 and 2 are the proportion backwards

Scoring: 1 point.

3. A drawing has a 35° angle. A scale copy is made at k = 3. What is the matching angle in the copy? *1 pt*

- ○ A. 105°
- ● **B. 35°**
- ○ C. 11.67°
- ○ D. 38°

B

Lengths scale; angles do not.

Scoring: 1 point.

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

42 cm

perimeter × k

b. Find the copy's area. *1 pt*

108 cm²

area × k² — 36, 40 and 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 8 cm². Its scale copy has area 72 cm². Find k, and explain why k is not 9. *1 pt*

k = 3. The area ratio is k², because area is two lengths each multiplied by k; 9 is k², so k is its square root.

The point needs k AND the reason (two lengths, so the ratio is k²). 'k = 3' alone is half the item; 'because you square root it' with no reason why is the rule without the reason.

Scoring: 1 point.

6. A floor plan is drawn at ¼ in : 1 ft. A room is 2 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

8 ft

the scale per 1 inch, then × — 1 in : 4 ft, 1 in : 2 ft, 1 in : 5 ft

b. Find the room's real area. *1 pt*

112 ft²

real dimensions first: 8 by 14, 6 by 10, 10 by 15 — the drawing's area × the rate once is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

4.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square metres. *1 pt*

About 9.30 m^2 (3.05 m by 3.05 m)

10 ft by 10 ft; 3.05 × 3.05 = 9.3025. The same number by the other route: 100 ft^2 × 0.305^2 = 9.3025. 30.5 (100 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 9 in? *1 pt*

22.86 cm

× the ratio whose 1 is inches

Scoring: 1 point.

10. 1 cm = 0.39 in. How many inches is 70 cm? *1 pt*

27.3 in (27.6 by ÷ 2.54)

× 0.39; the sheet's other ratio, ÷ 2.54, earns the point too when the work shows it. × 2.54 (177.8, 228.6, 279.4) is the inverted ratio.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 4 yd? *1 pt*

365.76 cm

a chain: 144 in, 180 in, 54 in, then × 2.54. The estimate (a yard is nearly a metre) catches 10.16, 12.7 and 3.81.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

44.08 lb

× the ratio whose 1 is the unit you have; the estimate says more pounds than kilograms

Scoring: 1 point.

13. 1 gal = 3.785 L. How many litres is 5 gal? *1 pt*

18.925 L

more litres than gallons

Scoring: 1 point.

14. 1 yd = 3 ft. A rug is 4 yd^2 in area. How many square feet is that? *1 pt*

- ○ A. 12 ft^2
- ○ B. 9 ft^2
- ● **C. 36 ft^2**
- ○ D. 24 ft^2

C

Scoring: 1 point.

15. Rate on 5 Nov: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

55.20 euros

× the rate

b. How many US dollars is 23 euros? *1 pt*

$25

÷ the rate; 21.16, 72.90 and 87.12 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

390 mi

the rate × the gallons

b. It starts a 1,000-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

3 tanks (1,000 ÷ 390 ≈ 2.56)

the part of a tank is still a tank; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 4 servings uses 1½ cups of rice.

a. How much rice for 10 servings? *1 pt*

3¾ cups

k from the yields (2.5, 1.5, 3), then the ingredient

b. With 6 cups on hand, how many servings can be made? *1 pt*

16 servings

recipes the pantry makes (4, 4, 5) × the yield; 15, 24 and 25 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

27 cups

k from the part you know (3, 4, 4), then the whole (9, 8, 6 parts); 12, 16 and 8 found one other part and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A 200-km drive. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how long? Explain. *1 pt*

Car A: 40 mi/h = 64.4 km/h > 60 km/h, so about 3.11 h against 3.33 h — about 13 minutes sooner.

The point needs the speeds in one unit, the choice, and the times (or the gap). 'Car B, 60 > 40' compared two units. A decision from two rates in different units appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.